Weihan Li

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Education

Georgia Institute of Technology, PhD in Machine Learning — Sept 2023 – Present

Georgia Institute of Technology, Master of Science in Computer Science — 2023 – 2025

- **Ph.D. Advisor**: Prof. Anqi Wu

Zhejiang University, Research Assistant — Sept 2020 – May 2023

- **Advisor**: Prof. Gang Pan

Zhejiang University, Bachelor in Computer Science and Technology — Sept 2016 – June 2020

Research Summary

My current research is in **robot learning for dexterous bimanual manipulation**: how multi-fingered robot hands can learn diverse manipulation skills from human demonstrations, and how skills learned in simulation can transfer reliably to real hardware. Previously, I studied how multiple brain regions communicate and how animal behavior can be understood at scale, with applications to brain disorders and brain-computer interfaces.

Research Experience: Robot Learning

Dexterous Bimanual Manipulation from Human Demonstrations — Aug 2026 – Present

Platform: bimanual 20-DoF dexterous hands (Wuji, Sharpa) on a Vega humanoid

- Turning human hand–object demonstrations into physically feasible robot demonstrations through kinematic and physics-based retargeting.
- Learning residual reinforcement-learning policies on top of the retargeted demonstrations to perform contact-rich manipulation robustly.
- Working toward sim-to-real transfer by aligning simulation with real-robot physics and evaluating policies against hardware-readiness criteria.

Research Experience: Machine Learning for Neuroscience and Behavior

Multimodal Animal Behavior Analysis from Pose and Video — Sep 2025 – Jul 2026

- Proposed a pose–video fusion framework in which the quantization residual of a pose-based motion encoder identifies the frames that need visual context, so that video features are added only where pose is insufficient.
- Applied the same model to supervised behavior classification and unsupervised behavior segmentation on three mouse datasets, outperforming pose- and video-based baselines on most tasks, with the largest gains in multi-animal social settings.
- Published at NeurIPS 2026.
- Co-developed BehaviorVLM, a finetuning-free approach to behavior understanding based on vision–language reasoning.

MBRILA: An Open-Source GP State-Space Framework for Multi-Region Brain Communication — Oct 2025 – May 2026

- Developed a general platform and algorithmic framework for Gaussian-process state-space (GP-SSM) modeling, in which any Gaussian-process model can be converted into an equivalent state-space form and trained with scalable, linear- or parallel-time inference.
- Provides shared building blocks (kernel-to-SSM conversion, multi-region latent dynamics, scalable inference) on which new GP-SSM methods can be built; MRM-GP and ADM are implemented as instances of the framework.
- Released under the MIT license at `github.com/BRAINML-GT/MBRILA`.

Modeling Time-Varying Communication Across Brain Regions (ADM) Aug 2024 – May 2025

- Proposed a state-space model that captures time-varying, directional interactions among multiple brain regions.
- Established a general correspondence between Gaussian processes and equivalent state-space models, allowing flexible and interpretable priors on inter-regional communication.
- Developed parallel-scan inference that reduces the time complexity from $O(T^3)$ to $O(\log T)$, enabling training on long multi-region recordings.
- Published at ICML 2025 as an **Oral** presentation.

Discovering Directional, Frequency-Specific Communication Across Brain Regions (MRM-GP) Sep 2023 – Jan 2024

- Approximated multi-output Gaussian processes with state-space models to infer directional communication across brain regions in linear time.
- Converted complex-valued spectral kernels into state-space dynamics, explicitly modeling frequency-specific coupling and phase delays between regions.
- Recovered feedforward and feedback interactions consistent with known anatomy on multi-region recordings.
- Published at ICML 2024.

Online Neural Sequence Detection with Hierarchical Dirichlet Point Process Jan 2022 – May 2022

- Developed a Bayesian nonparametric model that detects recurring neural sequences in spike trains and infers the number of sequence types from data.
- Derived an online inference algorithm based on particle filtering, detecting sequences as spikes arrive and scaling to long recordings.
- Published at NeurIPS 2022.

Publications

Learning When Visual Context Matters for Mouse Behavior Analysis **NeurIPS 2026**
Weihan Li, Jingyang Ke, Qinheng Pu, Yule Wang, Chengrui Li, Anqi Wu

MICE: Multi-animal Interaction Context Encoder — A Hierarchical Foundation Model for Mouse Behavior **NeurIPS 2026 Spotlight**
Yen-Shuo Su, ***Weihan Li***, Anqi Wu

QDMouse4M: A Multi-View 3D Mouse Spontaneous Behavior Dataset with Quantum-Dot Markers **NeurIPS 2026 E&D Track**
Jingyang Ke, Amartya Pradhan, Xueling Zhang, ***Weihan Li***, Anqi Wu, Jeffrey E. Markowitz

BehaviorVLM: Unified Finetuning-Free Behavioral Understanding with Vision-Language Reasoning preprint [link]
Jingyang Ke[*], ***Weihan Li***[*], Amartya Pradhan, Jeffrey Markowitz, Anqi Wu
[*]*Equal contribution; authors listed alphabetically.*

A Factorized Low-Rank RNN Framework for Uncovering Independent Neural Latent Dynamics and Connectivity **ICML 2026 Spotlight** [link]
Chengrui Li, Yunmiao Wang, Yule Wang, ***Weihan Li***, Dieter Jaeger, Anqi Wu

Uncovering Semantic Selectivity of Latent Groups in Higher Visual Cortex with Mutual Information-Guided Diffusion **ICLR 2026** [link]
Yule Wang, Joseph Yu, Chengrui Li, ***Weihan Li***, Dieter Jaeger, Anqi Wu

Learning Time-Varying Multi-Region Brain Communications via Scalable Markovian Gaussian Processes **ICML 2025 Oral** [link]
Weihan Li, Yule Wang, Chengrui Li, Anqi Wu

Exploring Behavior-Relevant and Disentangled Neural Dynamics with Generative Diffusion Models **NeurIPS 2024** [link]
Yule Wang, Chengrui Li, ***Weihan Li***, Anqi Wu

Multi-Region Markovian Gaussian Process: An Efficient Method to Discover Directional Communications Across Multiple Brain Regions **ICML 2024** [link]
Weihan Li, Chengrui Li, Yule Wang, Anqi Wu

A Differentiable Partially Observable Generalized Linear Model with Forward-Backward Message Passing **ICML 2024** [link]
Chengrui Li, ***Weihan Li***, Yule Wang, Anqi Wu

Forward χ^2 Divergence Based Variational Importance Sampling **ICLR 2024 Spotlight** [link]
Chengrui Li, Yule Wang, ***Weihan Li***, Anqi Wu

Online Neural Sequence Detection with Hierarchical Dirichlet Point Process **NeurIPS 2022** [link]
Weihan Li, Yu Qi, Gang Pan

Efficient Point-Process Modeling of Spiking Neurons for Neuroprosthesis **EMBC 2021** [link]
Weihan Li, Cunle Qian, Yu Qi, Yiwen Wang, Yueming Wang, Gang Pan

Technical Skills

Robotics & Simulation: Isaac Lab, MuJoCo, Motion Retargeting, Sim-to-Real Transfer

Machine Learning: PyTorch, Reinforcement Learning, Vision-Language Models

Languages: Python, C++

Teaching and Academic Services

Teaching Experience: TA for Georgia Tech CSE-8803 Statistical Machine Learning Models for Neural and Behavioral Data Analysis.

Conference Reviewer: NeurIPS 2023/2024/2025/2026, ICML 2024/2025/2026, ICLR 2024/2025/2026.